Matemàtiques
3r ESO

SOLUCIONS DE LA FITXA · UNITAT 6

Sistemes d'equacions

FentMates

Intenta resoldre cada exercici abans de mirar-ne la solució. Si no t'ix el mateix resultat, repassa els passos o consulta els apunts interactius de la unitat: hi trobaràs calculadores que resolen pas a pas qualsevol sistema pels tres mètodes i el dibuixen.

1 Equacions lineals amb dues incògnites

1. ★☆☆

a) sí **b)** no (x^2) **c)** no (xy)

d) sí **e)** no (una incògnita) **f)** sí

2. ★☆☆

a) $6+6=12$: sí **b)** $0+12=12$: sí **c)** $12+3=15$: no **d)** $-6+18=12$: sí

3. ★☆☆

a) $y=5-x$: $y=7,6,5,4,3$ **b)** $y=2x-1$: $y=-5,\ -3,\ -1,1,3$

4. ★★☆

a) $x=6+3y$: p. ex. $(6,0),(9,1),(3,-1)$ **b)** $y=5-2x$: p. ex. $(0,5),(1,3),(2,1)$

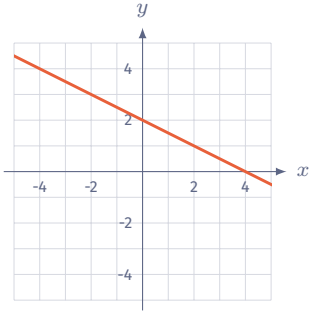
5. ★★☆

a) $-4+3k=8 \Rightarrow k=4$

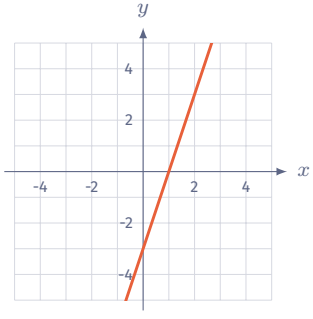
b) $10-2m=4 \Rightarrow m=3$

6. ★★☆

a) $(0,2),(2,1),(4,0)$



b) $(0,-3),(1,0),(2,3)$



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7. ★★★

a) $y = x + 3$: $3x + 2x + 6 = 21 \Rightarrow x = 3$: $\boxed{(3, 6)}$

b) Si $y = 0$, $x = 6$; si $x = 0$, $y = -3$: $\boxed{(6, 0) \text{ i } (0, -3)}$

2 Sistemes d'equacions i solucions

8. ★☆☆

a) $1 = 1$ i $5 = 5$: $\boxed{\text{sí}}$

b) $5 = 5$, però $2 + 2 = 4 \neq 3$: $\boxed{\text{no}}$

9. ★☆☆ $(1,2)$: $2 - 2 = 0$ ✗; $(2,1)$: 3 i 3 ✓; $(3,0)$: $6 \neq 3$ ✗. La solució és $\boxed{(2, 1)}$.

10. ★★☆ Per exemple, $\boxed{x + y = 3 \text{ i } x - y = -7}$ (hi ha infinites respostes).

11. ★★★ $3a - 4 = 5 \Rightarrow \boxed{a = 3}$; $12 - 2b = 6 \Rightarrow \boxed{b = 3}$

3 Mètode de substitució

12. ★☆☆

a) $x = 3y$
$(3y) + y = 8 \Rightarrow 3y + y = 8 \Rightarrow 4y = 8 \Rightarrow y = \frac{8}{4} = 2$
$x = 3 \cdot 2 = 6$
$\boxed{x = 6,\ y = 2}$

b) $y = 2x + 1$
$3x + (2x + 1) = 11 \Rightarrow 3x + 2x + 1 = 11 \Rightarrow 5x = 10 \Rightarrow x = \frac{10}{5} = 2$
$y = 2 \cdot 2 + 1 = 5$
$\boxed{x = 2,\ y = 5}$

13. ★☆☆

a) $x = 7 - y$
$2(7 - y) - y = 2 \Rightarrow 14 - 2y - y = 2 \Rightarrow -3y = -12 \Rightarrow y = \frac{-12}{-3} = 4$
$x = 7 - 4 = 3$
$\boxed{x = 3,\ y = 4}$

b) $x = y + 1$
$3(y + 1) + 2y = 13 \Rightarrow 3y + 3 + 2y = 13 \Rightarrow 5y = 10 \Rightarrow y = \frac{10}{5} = 2$
$x = 2 + 1 = 3$
$\boxed{x = 3,\ y = 2}$

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14. ★★☆

a) $y = 1 - 3x$
$5x - 2(1 - 3x) = 20 \Rightarrow 5x - 2 + 6x = 20 \Rightarrow 11x = 22 \Rightarrow x = \frac{22}{11} = 2$
$y = 1 - 3 \cdot 2 = -5$
$\boxed{x = 2,\ y = -5}$

b) $-y = -4 - 2x \Rightarrow y = 2x + 4$
$4x + 3(2x + 4) = 2 \Rightarrow 4x + 6x + 12 = 2 \Rightarrow 10x = -10 \Rightarrow x = \frac{-10}{10} = -1$
$y = 2 \cdot (-1) + 4 = 2$
$\boxed{x = -1,\ y = 2}$

15. ★★☆

a) $x = -5 - 4y$
$3(-5 - 4y) - 2y = 13 \Rightarrow -15 - 12y - 2y = 13 \Rightarrow -14y = 28 \Rightarrow y = \frac{28}{-14} = -2$
$x = -5 - 4 \cdot (-2) = 3$
$\boxed{x = 3,\ y = -2}$

b) $-y = 13 - 5x \Rightarrow y = 5x - 13$
$2x + 3(5x - 13) = 12 \Rightarrow 2x + 15x - 39 = 12 \Rightarrow 17x = 51 \Rightarrow x = \frac{51}{17} = 3$
$y = 5 \cdot 3 - 13 = 2$
$\boxed{x = 3,\ y = 2}$

16. ★★★

a) $2x = 12 - 3y \Rightarrow x = \frac{12-3y}{2}$
$5 \cdot \frac{12-3y}{2} - 4y = 7 \Rightarrow 5(12 - 3y) - 8y = 14 \Rightarrow 60 - 15y - 8y = 14 \Rightarrow -23y = -46 \Rightarrow y = \frac{-46}{-23} = 2$
$x = \frac{12-3\cdot 2}{2} = 3$
$\boxed{x = 3,\ y = 2}$

b) $-2y = -1 - 3x \Rightarrow y = \frac{3x+1}{2}$
$4x + 5 \cdot \frac{3x+1}{2} = 14 \Rightarrow 8x + 5(3x + 1) = 28 \Rightarrow 8x + 15x + 5 = 28 \Rightarrow 23x = 23 \Rightarrow x = \frac{23}{23} = 1$
$y = \frac{3\cdot 1+1}{2} = 2$
$\boxed{x = 1,\ y = 2}$

17. ★★★ $-3(3 - x) = -9 + 3x$: el -3 multiplica els dos termes. $y = 3 - x$
$2x - 3(3 - x) = 1 \Rightarrow 2x - 9 + 3x = 1 \Rightarrow 5x = 10 \Rightarrow x = \frac{10}{5} = 2$
$y = 3 - 2 = 1$
$\boxed{x = 2,\ y = 1}$

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4 Mètode d'igualació

18. ★☆☆

a) $y = x + 5, \quad y = 3x - 1$
$x + 5 = 3x - 1 \Rightarrow -2x = -6 \Rightarrow x = \frac{-6}{-2} = 3$
$y = 3 + 5 = 8$
$\boxed{x = 3,\ y = 8}$

b) $x = 2y - 3, \quad x = y + 1$
$2y - 3 = y + 1 \Rightarrow y = 4$
$x = 2 \cdot 4 - 3 = 5$
$\boxed{x = 5,\ y = 4}$

19. ★☆☆

a) $x = 6 - y, \quad x = y + 2$
$6 - y = y + 2 \Rightarrow -2y = -4 \Rightarrow y = \frac{-4}{-2} = 2$
$x = 6 - 2 = 4$
$\boxed{x = 4,\ y = 2}$

b) $y = 2x + 1, \quad y = 7 - x$
$2x + 1 = 7 - x \Rightarrow 3x = 6 \Rightarrow x = \frac{6}{3} = 2$
$y = 2 \cdot 2 + 1 = 5$
$\boxed{x = 2,\ y = 5}$

20. ★★☆

a) $y = 8 - 2x, \quad y = 11 - 3x$
$8 - 2x = 11 - 3x \Rightarrow x = 3$
$y = 8 - 2 \cdot 3 = 2$
$\boxed{x = 3,\ y = 2}$

b) $x = 10 - 3y, \quad x = 2y$
$10 - 3y = 2y \Rightarrow -5y = -10 \Rightarrow y = \frac{-10}{-5} = 2$
$x = 10 - 3 \cdot 2 = 4$
$\boxed{x = 4,\ y = 2}$

21. ★★☆

a) $-y = 7 - 3x \Rightarrow y = 3x - 7, \quad 2y = -x \Rightarrow y = \frac{-x}{2}$
$3x - 7 = \frac{-x}{2} \Rightarrow 2(3x - 7) = -x \Rightarrow 6x - 14 = -x \Rightarrow 7x = 14 \Rightarrow x = \frac{14}{7} = 2$
$y = 3 \cdot 2 - 7 = -1$
$\boxed{x = 2,\ y = -1}$

b) $y = -2 - 4x, \quad -y = -4 - 2x \Rightarrow y = 2x + 4$
$-2 - 4x = 2x + 4 \Rightarrow -6x = 6 \Rightarrow x = \frac{6}{-6} = -1$
$y = -2 - 4 \cdot (-1) = 2$
$\boxed{x = -1,\ y = 2}$

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22. ★★★

a) $2x = 7 - 3y \Rightarrow x = \frac{7-3y}{2}$, $3x = 2y + 4 \Rightarrow x = \frac{2y+4}{3}$
$\frac{7-3y}{2} = \frac{2y+4}{3} \Rightarrow 3(7-3y) = 2(2y+4) \Rightarrow 21 - 9y = 4y + 8 \Rightarrow -13y = -13 \Rightarrow y = \frac{-13}{-13} = 1$
$x = \frac{7-3\cdot 1}{2} = 2$
$\boxed{x = 2,\ y = 1}$

b) $-2y = 1 - 5x \Rightarrow y = \frac{5x-1}{2}$, $4y = 11 - 3x \Rightarrow y = \frac{11-3x}{4}$
$\frac{5x-1}{2} = \frac{11-3x}{4} \Rightarrow 2(5x-1) = 11 - 3x \Rightarrow 10x - 2 = 11 - 3x \Rightarrow 13x = 13 \Rightarrow x = \frac{13}{13} = 1$
$y = \frac{5\cdot 1-1}{2} = 2$
$\boxed{x = 1,\ y = 2}$

23. ★★★ Igualació: $\frac{x+1}{2} = \frac{2x-1}{3} \Rightarrow 3(x+1) = 2(2x-1) \Rightarrow 3x + 3 = 4x - 2 \Rightarrow x = 5$; $y = \frac{5+1}{2} = 3$.
$\boxed{x = 5,\ y = 3}$

5 Mètode de reducció

24. ★☆☆

a)
$$\begin{array}{rrcr} x & +y & = & 9 \\ x & -y & = & 3 \\ \hline 2x & & = & 12 \end{array} \Rightarrow 2x = 12 \Rightarrow x = \frac{12}{2} = 6$$
$6 + y = 9 \Rightarrow y = 3$
$\boxed{x = 6,\ y = 3}$

b)
$$\begin{array}{rrcr} 3x & +2y & = & 13 \\ x & -2y & = & -1 \\ \hline 4x & & = & 12 \end{array} \Rightarrow 4x = 12 \Rightarrow x = \frac{12}{4} = 3$$
$3 \cdot 3 + 2y = 13 \Rightarrow 2y = 4 \Rightarrow y = \frac{4}{2} = 2$
$\boxed{x = 3,\ y = 2}$

25. ★☆☆

a)
$$\begin{array}{rrrcr} & 2x & +y & = & 7 \\ (\cdot(-1)) & -2x & +3y & = & 5 \\ \hline & & 4y & = & 12 \end{array} \Rightarrow 4y = 12 \Rightarrow y = \frac{12}{4} = 3$$
$2x + 3 = 7 \Rightarrow 2x = 4 \Rightarrow x = \frac{4}{2} = 2$
$\boxed{x = 2,\ y = 3}$

b)
$$\begin{array}{rrcr} 5x & +3y & = & 11 \\ 2x & -3y & = & -4 \\ \hline 7x & & = & 7 \end{array} \Rightarrow 7x = 7 \Rightarrow x = \frac{7}{7} = 1$$
$5 \cdot 1 + 3y = 11 \Rightarrow 3y = 6 \Rightarrow y = \frac{6}{3} = 2$
$\boxed{x = 1,\ y = 2}$

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26. ★★☆

a)
$$\begin{array}{rrcr} & x & +2y & = & 4 \\ (\cdot 2) & 6x & -2y & = & 10 \\ \hline & 7x & & = & 14 \end{array} \Rightarrow 7x = 14 \Rightarrow x = \frac{14}{7} = 2$$

$3 \cdot 2 - y = 5 \Rightarrow -y = -1 \Rightarrow y = \frac{-1}{-1} = 1$

$\boxed{x = 2,\ y = 1}$

b)
$$\begin{array}{rrcr} (\cdot(-1)) & -4x & -3y & = & -10 \\ (\cdot 2) & 4x & +10y & = & 24 \\ \hline & & 7y & = & 14 \end{array} \Rightarrow 7y = 14 \Rightarrow y = \frac{14}{7} = 2$$

$2x + 5 \cdot 2 = 12 \Rightarrow 2x = 2 \Rightarrow x = \frac{2}{2} = 1$

$\boxed{x = 1,\ y = 2}$

27. ★★☆

a)
$$\begin{array}{rrcr} & 3x & +4y & = & 18 \\ (\cdot 2) & 10x & -4y & = & 8 \\ \hline & 13x & & = & 26 \end{array} \Rightarrow 13x = 26 \Rightarrow x = \frac{26}{13} = 2$$

$5 \cdot 2 - 2y = 4 \Rightarrow -2y = -6 \Rightarrow y = \frac{-6}{-2} = 3$

$\boxed{x = 2,\ y = 3}$

b)
$$\begin{array}{rrcr} (\cdot 3) & 6x & -15y & = & -3 \\ (\cdot(-2)) & -6x & -4y & = & -16 \\ \hline & & -19y & = & -19 \end{array} \Rightarrow -19y = -19 \Rightarrow y = \frac{-19}{-19} = 1$$

$2x - 5 \cdot 1 = -1 \Rightarrow 2x = 4 \Rightarrow x = \frac{4}{2} = 2$

$\boxed{x = 2,\ y = 1}$

28. ★★★

a)
$$\begin{array}{rrcr} (\cdot 2) & 8x & +6y & = & -2 \\ (\cdot(-3)) & -15x & -6y & = & -12 \\ \hline & -7x & & = & -14 \end{array} \Rightarrow -7x = -14 \Rightarrow x = \frac{-14}{-7} = 2$$

$5 \cdot 2 + 2y = 4 \Rightarrow 2y = -6 \Rightarrow y = \frac{-6}{2} = -3$

$\boxed{x = 2,\ y = -3}$

b)
$$\begin{array}{rrcr} (\cdot 2) & 12x & -10y & = & 14 \\ (\cdot(-3)) & -12x & +9y & = & -15 \\ \hline & & -y & = & -1 \end{array} \Rightarrow -y = -1 \Rightarrow y = \frac{-1}{-1} = 1$$

$4x - 3 \cdot 1 = 5 \Rightarrow 4x = 8 \Rightarrow x = \frac{8}{4} = 2$

$\boxed{x = 2,\ y = 1}$

29. ★★★ $(\cdot 2), (\cdot 3)$: $13x = 11 \Rightarrow \boxed{x = \frac{11}{13}}$; $(\cdot 3), (\cdot(-2))$: $13y = 10 \Rightarrow \boxed{y = \frac{10}{13}}$. Així evitem substituir una fracció.

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6 Mètode gràfic

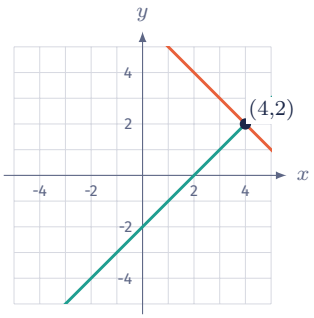
30. ★☆☆

a) Les rectes es tallen en $(1, 3)$: $\boxed{x = 1, y = 3}$

b) Les rectes es tallen en $(1, -1)$: $\boxed{x = 1, y = -1}$

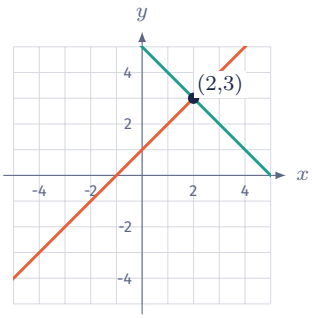
31. ★☆☆

a)



$\boxed{x = 4, y = 2}$

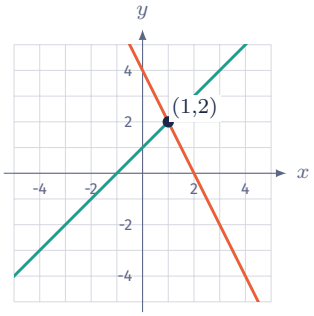
b)



$\boxed{x = 2, y = 3}$

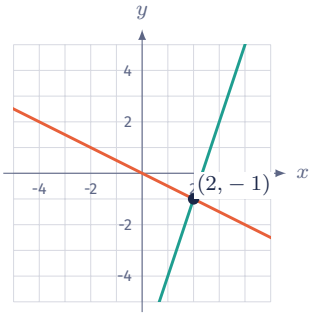
32. ★★☆

a)



$\boxed{x = 1, y = 2}$

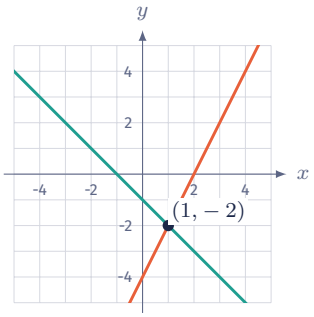
b)



$\boxed{x = 2, y = -1}$

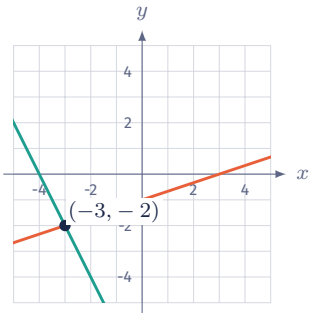
33. ★★☆

a)



$\boxed{x = 1, y = -2}$

b)

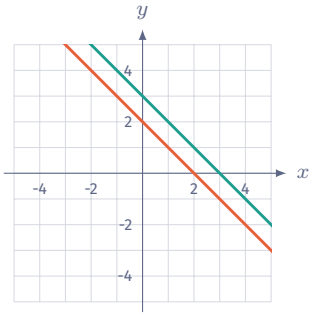


$\boxed{x = -3, y = -2}$

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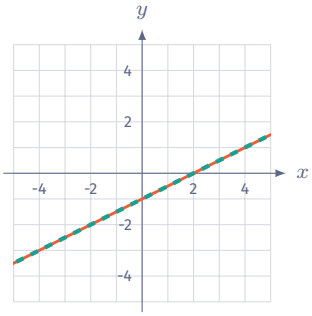
34. ★★★

a)



Rectes paral·leles: cap solució

b)



La mateixa recta: infinites solucions

7 Classificació dels sistemes

35. ★☆☆

a) $\frac{1}{2} = \frac{3}{6} = \frac{2}{4}$: SCI

b) $\frac{2}{4} = \frac{-1}{-2} \neq \frac{1}{3}$: SI

c) $\frac{1}{2} \neq \frac{1}{-1}$: SCD

d) $\frac{3}{-1} = \frac{-6}{2} = \frac{9}{-3}$: SCI

36. ★☆☆

a) una solució, compatible determinat

b) cap solució, incompatible

c) infinites solucions, compatible indeterminat

37. ★★☆

a)

$$\begin{array}{lrcr} (\cdot 2) & 2x & +2y & = & 6 \\ (\cdot(-1)) & -2x & -2y & = & -5 \\ \hline & 0 & = & 1 \end{array}$$

SI (cap solució)

b)

$$\begin{array}{lrcr} & x & -2y & = & 1 \\ (\cdot(-1)) & -x & +2y & = & -1 \\ \hline & 0 & = & 0 \end{array}$$

SCI (infinites solucions). Per exemple, $(1, 0)$, $(3, 1)$, $(-1, -1)$

38. ★★☆

a)

$$\begin{array}{lrcr} & 2x & +3y & = & 5 \\ (\cdot(-1)) & -2x & -3y & = & -5 \\ \hline & 0 & = & 0 \end{array}$$

SCI (infinites solucions). P. ex. $(1, 1)$, $(4, -1)$

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b)
$$\begin{array}{rcl} x - y = 2 \\ -x + y = 2 \\ \hline 0 = 4 \end{array}$$
SI (cap solució)

c)
$$\begin{array}{rcl} 3x + y = 2 \\ x - y = 2 \\ \hline 4x = 4 \end{array} \Rightarrow 4x = 4 \Rightarrow x = \frac{4}{4} = 1$$
$3 \cdot 1 + y = 2 \Rightarrow y = -1$
$\boxed{x = 1,\ y = -1}$ (SCD)

39. ★★☆

a) P. ex. $\boxed{x + y = 3}$ **b)** P. ex. $\boxed{4x - 2y = 6}$ **c)** P. ex. $\boxed{2x - y = 5}$

40. ★★★ $\frac{m}{3} = \frac{2}{1} \Rightarrow \boxed{m = 6}$; i aleshores $\frac{3}{1} = 3 \neq 2$: incompatible.
Si $m \neq 6$: compatible determinat.

41. ★★★ $\frac{2}{4} = \frac{a}{-6} = \frac{b}{10} = \frac{1}{2} \Rightarrow \boxed{a = -3,\ b = 5}$

8 Sistemes amb parèntesis i denominadors

42. ★☆☆

a) $2(x-1) + y = 7 \Rightarrow 2x + y = 9$
$x - (y - 3) = 3 \Rightarrow x - y = 0$
$y = 9 - 2x$
$x - (9 - 2x) = 0 \Rightarrow x - 9 + 2x = 0 \Rightarrow 3x = 9 \Rightarrow x = \frac{9}{3} = 3$
$y = 9 - 2 \cdot 3 = 3$
$\boxed{x = 3,\ y = 3}$

b) $3(x + y) - 2x = 10 \Rightarrow x + 3y = 10$
$2(x - y) + y = -1 \Rightarrow 2x - y = -1$
$x = 10 - 3y$
$2(10 - 3y) - y = -1 \Rightarrow 20 - 6y - y = -1 \Rightarrow -7y = -21 \Rightarrow y = \frac{-21}{-7} = 3$
$x = 10 - 3 \cdot 3 = 1$
$\boxed{x = 1,\ y = 3}$

43. ★☆☆

a) $\frac{x}{2} + y = 4 \Rightarrow \cdot 2:\ x + 2y = 8 \Rightarrow x + 2y = 8$
$x - \frac{y}{3} = 1 \Rightarrow \cdot 3:\ 3x - y = 3 \Rightarrow 3x - y = 3$
$x = 8 - 2y$
$3(8 - 2y) - y = 3 \Rightarrow 24 - 6y - y = 3 \Rightarrow -7y = -21 \Rightarrow y = \frac{-21}{-7} = 3$
$x = 8 - 2 \cdot 3 = 2$
$\boxed{x = 2,\ y = 3}$

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b) $\frac{x}{3}+\frac{y}{2}=2 \Rightarrow \cdot 6:\ 2x+3y=12 \Rightarrow 2x+3y=12$
$\frac{x}{2}-\frac{y}{4}=1 \Rightarrow \cdot 4:\ 2x-y=4 \Rightarrow 2x-y=4$

$$\begin{array}{rrrrr} & 2x & +3y & = & 12 \\ (\cdot(-1)) & -2x & +y & = & -4 \\ \hline & & 4y & = & 8 \end{array} \Rightarrow 4y=8 \Rightarrow y=\frac{8}{4}=2$$

$2x+3\cdot 2=12 \Rightarrow 2x=6 \Rightarrow x=\frac{6}{2}=3$

$\boxed{x=3,\ y=2}$

44. ★★☆

a) $\frac{x+1}{2}+\frac{y-1}{3}=3 \Rightarrow \cdot 6:\ 3(x+1)+2(y-1)=18 \Rightarrow 3x+2y=17$
$x=y-1$
$3(y-1)+2y=17 \Rightarrow 3y-3+2y=17 \Rightarrow 5y=20 \Rightarrow y=\frac{20}{5}=4$
$x=4-1=3$
$\boxed{x=3,\ y=4}$

b) $\frac{x-2}{3}=\frac{y+1}{4} \Rightarrow \cdot 12:\ 4(x-2)=3(y+1) \Rightarrow 4x-3y=11$
$-y=7-2x \Rightarrow y=2x-7$
$4x-3(2x-7)=11 \Rightarrow 4x-6x+21=11 \Rightarrow -2x=-10 \Rightarrow x=\frac{-10}{-2}=5$
$y=2\cdot 5-7=3$
$\boxed{x=5,\ y=3}$

45. ★★☆

$0{,}5x+0{,}2y=2 \Rightarrow 5x+2y=20$
$x=y-3$
$5(y-3)+2y=20 \Rightarrow 5y-15+2y=20 \Rightarrow 7y=35 \Rightarrow y=\frac{35}{7}=5$
$x=5-3=2$
$\boxed{x=2,\ y=5}$

46. ★★★

$\frac{2x-1}{3}-\frac{y+2}{2}=-1 \Rightarrow \cdot 6:\ 2(2x-1)-3(y+2)=-6 \Rightarrow 4x-3y=2$
$3(x-y)-2(x+1)=-6 \Rightarrow x-3y=-4$

$$\begin{array}{rrrrr} & 4x & -3y & = & 2 \\ (\cdot(-1)) & -x & +3y & = & 4 \\ \hline & 3x & & = & 6 \end{array} \Rightarrow 3x=6 \Rightarrow x=\frac{6}{3}=2$$

$4\cdot 2-3y=2 \Rightarrow -3y=-6 \Rightarrow y=\frac{-6}{-3}=2$
$\boxed{x=2,\ y=2}$

47. ★★★

a) Igualació (la y ja està aïllada): $y=3x-2, \quad y=x+4$
$3x-2=x+4 \Rightarrow 2x=6 \Rightarrow x=\frac{6}{2}=3$
$y=3\cdot 3-2=7$
$\boxed{x=3,\ y=7}$

b) Substitució (x té coeficient 1): $x=9-5y$
$2(9-5y)-3y=5 \Rightarrow 18-10y-3y=5 \Rightarrow -13y=-13 \Rightarrow y=\frac{-13}{-13}=1$
$x=9-5\cdot 1=4$
$\boxed{x=4,\ y=1}$

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c) Reducció (els coeficients de la y són oposats):
$$\begin{array}{rrrcr} (\cdot(-1)) & -4x & -3y & = & -6 \\ (\cdot 2) & 4x & -2y & = & 16 \\ \hline & & -5y & = & 10 \end{array} \Rightarrow -5y = 10 \Rightarrow y = \frac{10}{-5} = -2$$
$2x - (-2) = 8 \Rightarrow 2x = 6 \Rightarrow x = \frac{6}{2} = 3$
$\boxed{x = 3,\ y = -2}$

9 Problemes amb sistemes d'equacions

48. ★☆☆ $\begin{cases} x + y = 56 \\ x - y = 12 \end{cases} \Rightarrow 2x = 68 \Rightarrow x = 34,\ y = 22$: **34 i 22**

49. ★☆☆ $\begin{cases} x = 3y \\ x + y = 48 \end{cases} \Rightarrow 4y = 48 \Rightarrow y = 12,\ x = 36$: **36 i 12**

50. ★☆☆ $\begin{cases} x + y = 40 \\ 2x + 4y = 112 \end{cases} \Rightarrow 2y = 32 \Rightarrow y = 16,\ x = 24$: **24 gallines i 16 conills**

51. ★☆☆ $\begin{cases} 3x + 2y = 8 \\ x + 3y = 5 \end{cases} \Rightarrow x = 5 - 3y \Rightarrow 15 - 7y = 8 \Rightarrow y = 1,\ x = 2$: **pomes 2 €/kg, peres 1 €/kg**

52. ★★☆ $\begin{cases} x + y = 22 \\ 2x + 0{,}5y = 26 \end{cases} \Rightarrow 1{,}5x = 15 \Rightarrow x = 10,\ y = 12$: **10 de 2 € i 12 de 50 cèntims**

53. ★★☆ $\begin{cases} x + y = 85 \\ 7x + 4y = 445 \end{cases} \Rightarrow 3x = 105 \Rightarrow x = 35,\ y = 50$: **35 d'adult i 50 de xiquet**

54. ★★☆ $\begin{cases} x = 4y \\ x + 5 = 3(y + 5) \end{cases} \Rightarrow 4y + 5 = 3y + 15 \Rightarrow y = 10$: **mare 40 anys, fill 10 anys**

55. ★★☆ $\begin{cases} 2x + 2y = 46 \\ x - y = 7 \end{cases} \Rightarrow x + y = 23 \Rightarrow 2x = 30$: **base 15 m, altura 8 m**

56. ★★☆ $\begin{cases} x + y = 30 \\ 2x - y = 39 \end{cases} \Rightarrow 3x = 69 \Rightarrow x = 23,\ y = 7$: **23 encerts (i 7 errades)**

57. ★★☆ $\begin{cases} x + y = 28 \\ y + 2 = x \end{cases} \Rightarrow 2y + 2 = 28 \Rightarrow y = 13$: **15 xics i 13 xiques**

58. ★★★ $\begin{cases} x + y = 60 \\ 9x + 6y = 420 \end{cases} \Rightarrow 3x = 60 \Rightarrow x = 20,\ y = 40$: **20 kg de 9 € i 40 kg de 6 €**

59. ★★★ $\begin{cases} x + y = 11 \\ 10y + x = 10x + y - 27 \end{cases} \Rightarrow \begin{cases} x + y = 11 \\ x - y = 3 \end{cases} \Rightarrow x = 7,\ y = 4$: **74**

60. ★★★ $\begin{cases} 3x + 3y = 75 \\ x - y = 5 \end{cases} \Rightarrow x + y = 25 \Rightarrow x = 15,\ y = 10$: **15 km/h i 10 km/h**

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61. ★★★ $\begin{cases} 3(x+1) = 2(y+1) \\ 2(x-1) = y-1 \end{cases} \Rightarrow \begin{cases} 3x - 2y = -1 \\ 2x - y = 1 \end{cases} \Rightarrow x = 3,\ y = 5$: $\boxed{\frac{3}{5}}$

62. ★★★ $\begin{cases} x + y = 50 \\ 2x + y = 87 \end{cases} \Rightarrow x = 37,\ y = 13$: 37 dobles i 13 individuals

63. ★★★ $\begin{cases} 2x + y = 40 \\ x = y + 5 \end{cases} \Rightarrow 3y + 10 = 40 \Rightarrow y = 10$: costats iguals 15 cm, base 10 cm

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